

N83 retained-domain gates and K1182 storage

2026-10-10 | Public known-answer research | Frozen source 85e14930ff0aab0d50e8fb12ee7ca382e5e2f8e2

The minimum complete cold index-calculus runtime remains unmeasured. These receipts establish exact storage and bounded model/search admission, not a factor-base winner.

Exact workload

icv1-f2m83-tm6151469093347-debefd74: $y^2 + xy = x^3 + 1$ over $z^{83} + z^{45} + z^2 + z + 1$. Subgroup order 2417851639230796216685689; cofactor 4. The 53-bit diagnostic arm is a separate workload.

Owner budget: 7,200 seconds total, including 3,593.092994294 seconds charged before this extension. The extension ceiling is 3,606.907005706 seconds. The source-matched cross-build charged 872 seconds. Native guarded runs use one pinned CPU, 4 GiB, zero swap, no network, read-only inputs/root and fixed wall caps. The shared arm64 VM runs the x86-64 static worker under emulation; elapsed values are budget diagnostics only.

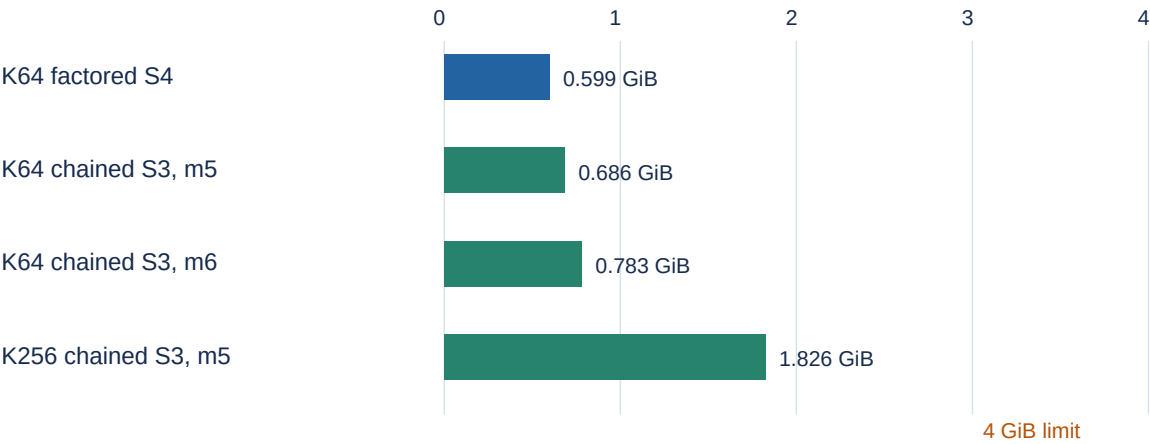
Observed gates

Gate	Measured observation	Outcome
K64 factored S4	127,404 SAT variables; 1,490,138 clauses; 1,074 XOR rows; 642,768,896-byte cgroup peak	Construction PASS; search false
K64 chained S3, m5	84,743 variables; 2,100,012 clauses; 1,851,925 finite-domain clauses; 736,542,720-byte peak	Construction PASS
K64 chained S3, m6	105,908 variables; 2,532,398 clauses; 2,222,310 finite-domain clauses; 841,170,944-byte peak	Construction PASS
K256 chained S3, m5	84,743 variables; 7,443,537 clauses; 7,195,450 finite-domain clauses; 1,960,288,256-byte peak	Construction PASS after separate 8-million-clause gate
K1182 base	196,212 signed-Frobenius records in 1,182 columns; 5,956,114 compressed bytes	Construction, generic replay, S3 round-trip PASS

Each model has one memory sample, without an uncertainty estimate. Measured finite-domain models and historical algebraic source bounds have different clause scopes.

Capacity and stored-object coverage

Measured cgroup peak memory - one run per model



Construction admission only; shared-host timings are not compared.

All point records verified

K1182 contains 196,212 signed-Frobenius point records in 1,182 columns. Generic multi-limb BinaryCurve replay checked every point and representative against the native Gf2_128 producer on the same host/repository. The compressed object is 5,956,114 bytes.

Point-set BLAKE3: 429cde4bc514bf1026e4993e269192f858cb7410ade777178a1c0db7dbf0001a

Compressed BLAKE3: 76e1cbe4ea5a9d26e5b24a2d27f9401271dc471846529190a0416d8e09837a92

The native uploader downloaded the S3 object and checked the compressed bytes. Full objects stay outside Git; content-addressed receipts are committed. Storage coverage is 55 constructed objects, while external-host arithmetic replay is pending.

Bounded search and retained failures

Case	Final observation
K64 m5	One SAT call; 10,000-conflict limit reached; zero models, relations and rank rows. UNKNOWN_solver_cap.
K64 m6	Docker daemon EOF; container later exited 255 without worker summary. PRODUCER_FAILURE; no solver verdict. Original cleanup field invalid, corrected recovery audit PASS.
K256 m5, 3M gate	Exact domain required 7,195,450 clauses; UNKNOWN_domain_clause_cap before model construction.
K256 m5, 8M gate	Construction and zero-trial preflight PASS. One SAT call; 10,000-conflict limit reached; zero models, relations and rank rows. UNKNOWN_solver_cap.

The first K1182 launch failed at container Git ownership admission; a process-local safe.directory fixed the fresh retry. A host build receipt wrapper failed after compilation on a reserved zsh variable; the clean retry is retained. The six-summand Docker EOF is an infrastructure producer failure. Its original cleanup flag is invalid; the exact exited container state was captured and removal independently verified. No solver verdict follows from that attempt.

Coverage against the requested objective

Requirement	Status and exact gap
Review prior art	Source-pinned study inventory and current-source integration audit retained.
Every combination	57,024,000 finite tuples addressable; most unexecuted. Values outside the grid remain unsearched.
All named techniques	Splitting, symmetry, WDSat, Gray, FES, double large primes and Frobenius in explicit axes and gates. Most backend combinations lack N83 execution.
Store factor bases in S3	54 historical objects plus one K1182 object with generic replay and verified compressed-byte round-trip.
Best complete cold runtime	Pending natural rank, complete relation collection, full-width solve, target verification, all charged phases and matched control.

What remains open

The finite v1/v2 design addresses 57,024,000 tuples, most unexecuted. WDSat/FES width and system-shape gates, a useful large-prime partial producer, full-width sparse LA, natural independent rank, individual logarithm, group verification and fully charged matched cold comparisons remain open. A K1182 construction-only chained-S3 entry point also needs extension before that larger object can enter its search gate. Censored UNKNOWN outcomes are not refutations.

Canonical graph treatment

The index-calculus scoreboard receives a source-linked capacity/storage coverage panel. docs/ic/progress-timeline.json, the cold leaderboard, docs/curves and docs/performance-gains were checked. No admitted runtime ratio, exponent or curve-map result is available for their plotted series. No new algebraic identity is claimed.

Evidence locations

Frozen protocol, commands, source/input hashes, image IDs, worker/outer JSON, logs, container states, recovery audit, tests and this PDF source are in research/koblitz_n83_factor_base_sweep_20261008/verification/two-hour-extension-20261010/. The durable object is in s3://crypto-autoresearcher/factor-bases/icv1/etc/koblitz_n83_factor_base_sweep_20261008/v2-size-frontier/a0/objects/ with filename equal to the compressed digest plus .jsonl.gz. Review PR #1599 for the same-branch source and evidence change.